MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

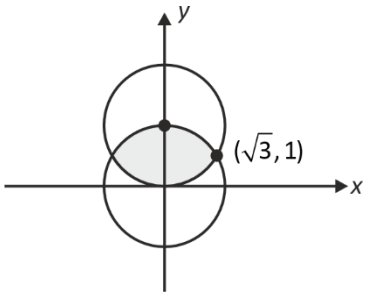
1. The area (in square units) between the curves $x^2 + y^2 = 4$ and $x^2 + (y-2)^2 = 4$ is

(1) $\frac{8\pi}{3} - 2\sqrt{3}$ (2) $\frac{8\pi}{3} + \sqrt{3}$

(3) $\frac{4\pi}{3} - 2\sqrt{3}$ (4) $\frac{4\pi}{3} + 2\sqrt{3}$

Answer (1)

Sol.



$$A = 2\int_0^{\sqrt{3}} \left(\sqrt{4-x^2} - \left(2 - \sqrt{4-x^2}\right)\right)dx$$

$$= 2\left(\left[x\sqrt{4-x^2} + 4\sin^{-1}\left(\frac{x}{2}\right)\right|_0^{\sqrt{3}} - 2(\sqrt{3})\right]\right)$$

$$= 4\left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right) \text{sq. units} = \frac{8\pi}{3}$$

2. Number of ways to distribute 6 identical oranges among 4 persons such that each gets at least one orange is

(1) 8

(2) 10

(3) 12

(4) 13

Answer (2)

Sol. $x_1 + x_2 + x_3 + x_4 = 6, \quad x_i \geq 1$

$\Rightarrow x_1' + x_2' + x_3' + x_4' = 2 \quad x_i' \geq 0$

$\Rightarrow$ Number of ways $= {}^{2+4-1}C_{4-1}$

$= {}^5C_3 = 10$ ways

3. The minimum value of $\cos^2\theta + 6\sin\theta\cos\theta + 3\sin^2\theta + 3$

(1) -1 (2) 1

(3) $5 + \sqrt{10}$ (4) $5 - \sqrt{10}$

Answer (4)

Sol. $\cos^2\theta + 6\sin\theta\cos\theta + 3\sin^2\theta + 3$

$4 + 3\sin 2\theta + 2\sin^2\theta$

$4 + 3\sin 2\theta + 1 - \cos 2\theta$

$E = 5 + 3\sin 2\theta - \cos 2\theta$

$E_{\min} = 5 - \sqrt{10}$

4. Let $A = \{1, 2, 3, \ldots, 9\}$; xRy iff $x - y$ is multiple of 3.

S_1 : Number of elements in R is 36

S_2 : R is equivalence relation

(1) S_1 & S_2 both are correct

(2) S_1 is correct but S_2 is not correct

(3) S_2 is correct but S_1 is not correct

(4) S_1 & S_2 both are incorrect

Answer (3)

Sol. S_1 : $(x, y) \equiv (3k_1, k_2)$ OR $(3k_1 + 1, 3k_2 + 1)$ OR $(3k_1 + 2, 3k_2 + 2)$

$\equiv 3 \times 3 + 3 \times 3 + 3 \times 3$

$= 27$ elements

S_1 is incorrect.

R is reflexive, symmetric as well as transitive relation.

as $(x, x) \in R \qquad \forall x \in A$

$(x, y) \in R \rightarrow (y, x) \in R$

$(x, y) \in R$ and $(y, z) \in R \Rightarrow (x, z) \in R$.



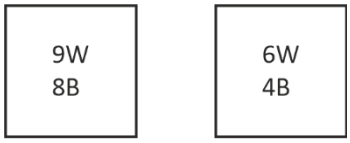
5. Let there are 2 bags *A* and *B*. Bag A has 9 white, 8 black balls and bag *B* has 6 white balls, 4 black balls. From a bag *B*, a ball is randomly selected and put into the bag *A*. Now, a ball is randomly selected from bag *A*. If the probability that selected ball is white is $\frac{p}{q}$, $gcd(p, q) = 1$, then $p + q$ is equal to

(1) 21 (2) 23

(3) 22 (4) 24

Answer (2)

Sol.



A : probability of getting white from bag A

B_w : probability of getting white from bag B

B_b : probability of getting black from bag B

$$P\left(\frac{A}{B_w \cup B_b}\right) = P(B_w)P\left(\frac{A}{B_w}\right) + P(B_b)P\left(\frac{A}{B_b}\right)$$

$$= \frac{6}{10}\times\frac{10}{18} + \frac{4}{10}\times\frac{9}{18}$$

$$= \frac{60+36}{180} = \frac{96}{180} = \frac{8}{15}$$

6. The value of $\int \frac{dx}{(4x+6)\sqrt{4x^2+12x+7}}$ is equal to

(1) $\frac{\sqrt{2}}{8}\tan^{-1}\left(\frac{4x^2+12x+7}{\sqrt{2}}\right)+c$

(2) $\frac{\sqrt{2}}{8}\tan^{-1}(4x^2+12x+7)+c$

(3) $\frac{1}{2}\tan^{-1}\left(\frac{4x^2+12x+7}{2}\right)+c$

(4) $\frac{\sqrt{2}}{4}\tan^{-1}\left(\frac{4x^2+12x+7}{\sqrt{2}}\right)+c$

Answer (1)

Sol. $I = \int \frac{(2x+3)}{2.(2x+3)^2\sqrt{4x^2+12x+7}}dx$

Let $4x^2+12x+7 = t^2$

$\therefore \quad (2x+3)dx = \frac{2tdt}{4}$

$\therefore \quad I = \int \frac{2t\,dt}{8.(4x^2+12x+9)t} = \int \frac{dt}{4(t^2+2)}$

$$= \frac{1}{4}\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{t}{\sqrt{2}}\right)$$

$\therefore \quad I = \frac{\sqrt{2}}{8}\tan^{-1}\left(\frac{4x^2+12x+7}{\sqrt{2}}\right)+c$

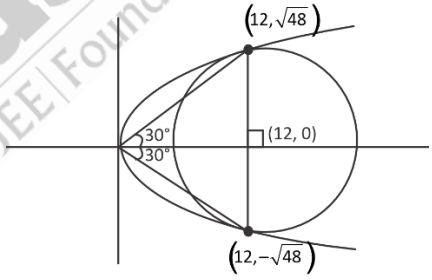
7. Let *A* and *B* are points on parabola $y^2 = 4x$ such that *OAB* is an equilateral triangle such that *O* is the vertex of parabola and *AB* is perpendicular to axis of parabola, then the minimum distance of circle as *A* and *B* as diametric points from the point *O* is

(1) $4\left(3+\sqrt{3}\right)$ (2) $4\left(3-\sqrt{3}\right)$

(3) $4\left(2-\sqrt{3}\right)$ (4) $4\left(2+\sqrt{3}\right)$

Answer (2)

Sol.



$\Rightarrow$ circle is

$$(x-12)^2 + \left(y-\sqrt{48}\right)\left(y+\sqrt{48}\right) = 0$$

$\Rightarrow (x-12)^2 + y^2 = 48 \quad \Rightarrow$ radius = $4\sqrt{3}$

Minimum distance is $12-\sqrt{48}$

$= 4\left(3-\sqrt{3}\right)$



8. The mean and variance of following data is μ and 19, respectively.

Class interval	4 – 8	8 – 12	12 – 16	16 – 20
***f* (frequency)**	3	λ	4	7

The value of $\lambda + \mu$ is

(1) 19 (2) 20

(3) 13 (4) 17

Answer (1)

Sol. $\mu = \frac{3\times b+\lambda\times 10+4\times 14+7\times 18}{14\times\lambda} = \frac{10\lambda+200}{14+\lambda}$

$$19=\sigma^2=\frac{3\times 6^2+\lambda(10)^2+4(14)^2+7(18)^2}{14+\lambda}-\left(\frac{10\lambda+200}{14+\lambda}\right)^2$$

$\Rightarrow \lambda = 6$

$\mu = 13$

$\lambda + \mu = 19$

9. If $\vec{a},\vec{b}$ and $\vec{c}$ are such that $\vec{a}\times\vec{b}=2(\vec{a}\times\vec{c})$ and angle between $\vec{b}$ and $\vec{c}$ is 60°, if $|\vec{a}|=1,|\vec{b}|=4,|\vec{c}|=3$, then $|\vec{a}.\vec{c}|$ is

(1) $\frac{3}{\sqrt{7}}$ (2) $\frac{5}{\sqrt{7}}$

(3) $\frac{6}{\sqrt{7}}$ (4) $\frac{9}{\sqrt{7}}$

Answer (3)

Sol. $\because \ \vec{a}\times\vec{b}=2\vec{a}\times\vec{c}$

$\therefore \ \vec{a}\times(\vec{b}-2\vec{c})=0$

$\therefore \ \vec{b}-2\vec{c}=\lambda\vec{a}$ ...(1) (where $\lambda\in R$)

and $\vec{b}.\vec{c}=|\vec{b}||\vec{c}|\cos 60°=6$ ... (2)

Squaring both sides of equation (1) we get :

$|\vec{b}|^2+4|\vec{c}|^2-4\vec{b}.\vec{c}=\lambda^2|\vec{a}|^2$

$\therefore \ \lambda^2=16+36-24=28$

from (1) : $\vec{b}.\vec{c}-2\vec{c}.\vec{c}=\lambda\vec{a}.\vec{c}$

$\therefore |\vec{a}.\vec{c}|=\frac{1}{\pm\sqrt{28}}(6-18)=\frac{12}{\sqrt{28}}=\frac{6}{\sqrt{7}}$

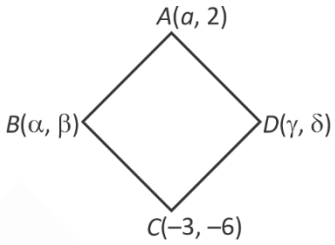
10. Let *A*, *B*, *C*, *D* be rhombus with $A(a, 2)$, $B(\alpha, \beta)$, $C(-3, -6)$ and $D(\gamma, \delta)$, then the value of $|\alpha + \beta + \gamma + \delta|$ is if *AC* is parallel to $y = 2x + 14$ is

(1) 0 (2) 1

(3) 6 (4) 9

Answer (3)

Sol.



$\because$ *AC* is parallel to $y = 2x + 14$

$\Rightarrow \frac{8}{a+3}=2\Rightarrow a=1$

Now $\frac{1-3}{2}=\frac{\alpha+\gamma}{2}\Rightarrow\alpha+\gamma=-2$

$\frac{2-6}{2}=\frac{\beta+\delta}{2}\Rightarrow\beta+\delta=-4$

$|\alpha + \beta + \gamma + \delta| = 6$

11.

12.

13.

14.

15.

16.

17.

18.

19.

20.



SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. If $A=\begin{bmatrix}0 & 2 & -3\\ -2 & 0 & -1\\ 3 & 1 & 0\end{bmatrix}$ and if $B(I-A)=(I+A)$, where I is identity matrix of order 3, then trace of (BB^T) is

Answer (03.00)

Sol. $B(I-A)=(I+A)$

Since A is skew symmetric matrix

$\Rightarrow A^T=-A$ since, $B^T(I-A^T)=(I+A^T)$

$\Rightarrow B^T(I+A)=(I-A)$

$\Rightarrow BB^T(I+A)=B(I-A)=(I+A)$

$\Rightarrow BB^T(I+A)(I+A)^{-1}=(I+A)(I+A)^{-1}$

$=I_{3\times3}$

$\Rightarrow \text{Trace}(BB^T)=3$

22. S = No. of 4- digit numbers *abcd* where product of digits is 20

P = No. of 5-digit number *abcde* where product of digits is 20, then

$S+P$ is equal to

Answer (74)

Sol. $\{a, b, c, d\}$

Digits : 5, 4, 1, 1 $\Rightarrow$ No. of arrangement $\Rightarrow \frac{4!}{2!}=12$

5, 2, 2, 1 $\Rightarrow$ No. of arrangement $\Rightarrow \frac{4!}{2!}=12$

$\Rightarrow S = 12 + 12 = 24$

$\{a, b, c, d, e\}$

Digits : 5, 4, 1, 1, 1 $\Rightarrow$ No. of arrangement $\Rightarrow \frac{5!}{3!}=20$

5, 2, 2, 1, 1 $\Rightarrow$ No. of arrangement $\Rightarrow \frac{5!}{2!2!}=30$

$\Rightarrow P = 20 + 30 = 50$

$S + P = 74$

23. Let S be the set defined as

$S=\left\{x:\int_0^x t^2\sin(t-x)dt=x^2 \text{ and } x\in[0, 1000]\right\}$, then number of elements in S is

Answer (1)

Sol. Let $I=\int_0^x t^2\sin(t-x)dt$

$=\int_0^x (x-t)^2\sin((x-t)-x)dt$

$=\int_0^x (x-t)^2\sin(-t)dt$

$=-x^2\int_0^x(\sin t)dt+2x\int_0^x t\sin t dt-\int_0^x t^2\sin t dt$

$=-x^2(-\cos t)|_0^x+2x(\sin t-t\cos t)|_0^x$

$-(2t\sin t+(2-t^2)\cos t)|_0^x$

$=x^2\cos x-x^2+2x(\sin x-x\cos x)$

$-2x\sin x+(x^2-2)\cos x+2$

$=x^2\cos x-x^2+2x\sin x-2x^2\cos x-2x\sin x$

$+x^2\cos x-2\cos x+2$

$=-x^2-2\cos x+2$

$\Rightarrow -x^2-2\cos x+2=x^2\Rightarrow 2(1-\cos x)=2x^2$

$\Rightarrow 2\sin^2\frac{x}{2}=x^2\Rightarrow x=0$ only solution

24. The number of solutions of the equation $\log_{(x-3)}(2x^2-7x+3)=2\log_{(2x-1)}(x-3)$ is equal to

Answer (1)

Sol. $\log_{(x-3)}(2x^2-7x+3)=2\log_{(2x-1)}(x-3)$

or,

$\log_{(x-3)}(x-3)(2x-1)=2\log_{(2x-1)}(x-3)$

or,

$1+\log_{(x-3)}(2x-1)=2\log_{(2x-1)}(x-3)$

Let $\log_{(x-3)}(2x-1)=y$

$\therefore 1+y=\frac{2}{y}$

Or, $y^2+y-2=0$

$\therefore \log_{(x-3)}(2x-1)=1,-2$



$\therefore \quad 2x-1=x-3$

$\therefore \quad x=-2$ not acceptable.

and $2x-1=(x-3)^{-2}$

$\therefore (2x-1)(x-3)^2=1$

By solving we get :

$(2x-5)(x^2-4x+2)=0$

$\therefore \quad x=\frac{5}{2}, 2+\sqrt{2}, 2-\sqrt{2}$

Only $x=2+\sqrt{2}$ satisfy the equations

$\therefore$ Number of solutions = 1.

25. If a sequence <a_n> satisfy the relation $\sum_{k=1}^{n} a_k = \alpha n^2 + \beta n$, and a_{10} = 56 and $a_6 = 2a$, then $(\alpha + \beta)$ is equal to

Answer (20)

Sol. $\sum_{k=1}^{n} a_k = \alpha n^2 + \beta n$

$a_{10} = 56 = a + 9d$

$a_6 = 2a_1 \Rightarrow a + 5d = 2a$

$= a + 5d$

$\Rightarrow 14d = 56 \Rightarrow d = 4$ and $a = 20$

$\sum_{k=1}^{n} a_k = \frac{n}{2}\left(2\times 20 + (n-1)4\right)$

$= 20n + 2(n)(n-1)$

$= 2n^2 + 18n$

$\Rightarrow \alpha = 2, \beta = 18 \Rightarrow \alpha + \beta = 20$

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